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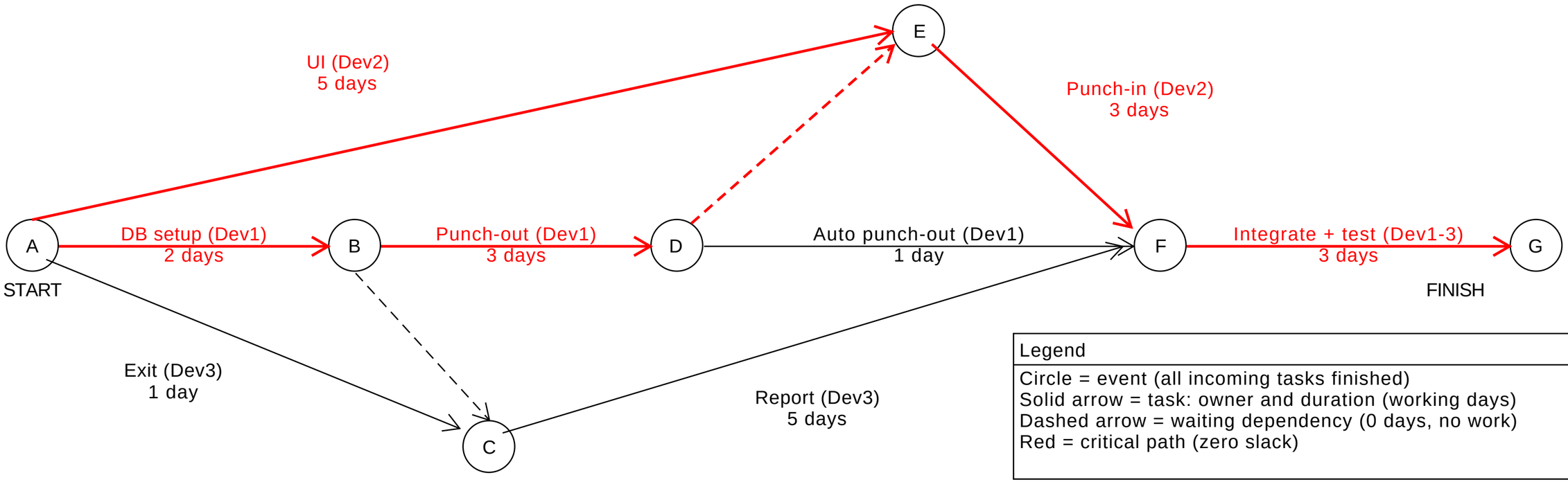


Legend

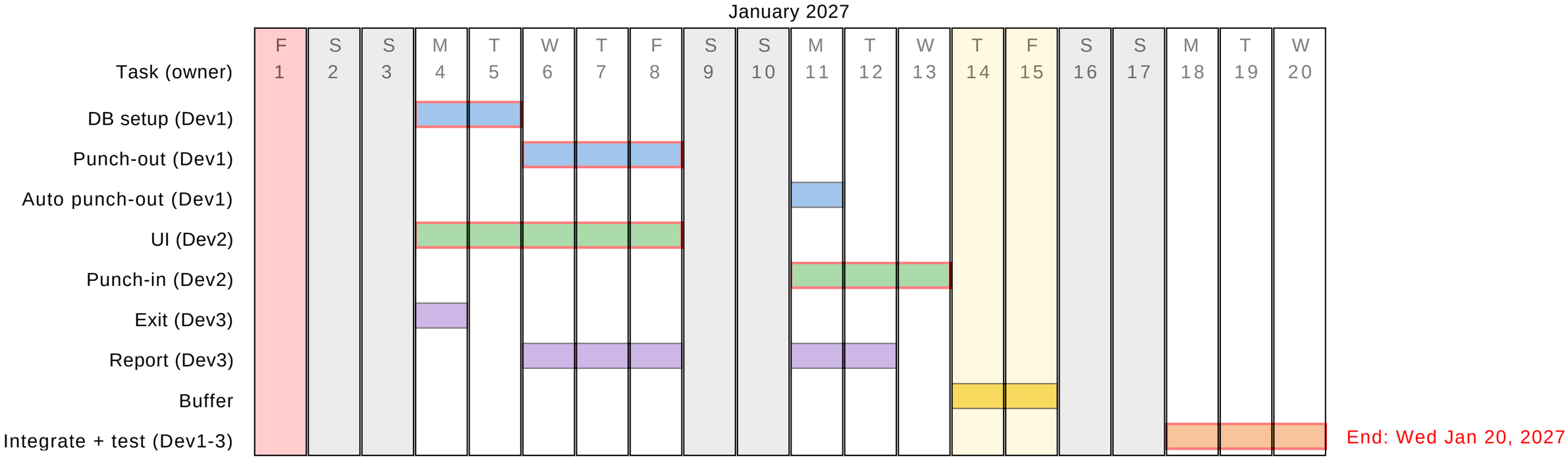
Circle = event (all incoming tasks finished)
Solid arrow = task: owner and duration (working days)
Dashed arrow = waiting dependency (0 days, no work)
Red = critical path (zero slack)

Node table

Edge	Dev	Days	Task	Pre-conditions
A-B	Dev1	2	DB setup	Spec and design approved
A-E	Dev2	5	UI (all screens)	Spec and design approved
A-C	Dev3	1	Exit + "Are you sure?"	Spec and design approved
B-D	Dev1	3	Punch-out	DB schema ready
D-E	-	0	(waiting)	Punch-in reuses punch-out
B-C	-	0	(waiting)	Report reads the DB
E-F	Dev2	3	Punch-in	Punch-out done, Dev2 free
D-F	Dev1	1	Auto punch-out 5:30 PM	Punch-out done
C-F	Dev3	5	Jobs Completed report	DB ready, Dev3 free
F-G	Dev1-3	3	Integrate + test	All tasks done

Q2: The project takes 11 working days, set by the critical path DB (2) + punch-out (3) + punch-in (3) + integrate and test (3).

Q3: The developers must be paid for 29 dev-days (20 for single-developer tasks plus 3 days x 3 developers for integration), which is 29 x 8 h = 232 hours.



Legend
Blue bar = Dev1 Green bar = Dev2 Purple bar = Dev3 Orange bar = all 3 devs Red outline = critical path task Grey column = weekend Red column = holiday Yellow column / bar = buffer day
Buffer: 2 days before Integrate + test, the riskiest step, because it is the first time all three devs' work is combined.

Q4: Work starts on Monday, January 4, 2027, because January 1 is New Year's Day and January 2 and 3 are a weekend. Two buffer days (January 14 and 15) come before integration, the riskiest step.

Q5: The project ends on Wednesday, January 20, 2027.

Q6: The Gantt chart allocates 9 extra days: 6 weekend days, 1 holiday and 2 buffer days.

Type	Days	Dates
Weekend	6	Jan 2, 3, 9, 10, 16, 17
Holiday	1	Jan 1 (New Year's Day)
Buffer	2	Jan 14, 15
Grand total	**9**	